MPWM and ADC Setting Guidelines for FOC Motor Control

AN Rev. 1.02

Introduction

This document describes how to set up the Motor Pulse Width Modulation (MPWM) and Analog to Digital Converter (ADC) used in the Field Oriented Control (FOC) algorithm, which is applied to the motor application of A34M41x, ABOV's ARM 32-bit Cortex-M4F core-based MCU.

Table 1 provides the list of microcontroller devices to which this document applies.

Table 1. Applicable devices

Sub-family	Part name
A34M41x	A34M418YL, A34M418VL, A34M418RL, A34M416VL, A34M416RL, A34M414VL, A34M414RL

Features and functions of each MCU product in Table 1 are provided in the appropriate User's manual. The main features of A34M41x product are summarized in Table 7 of the Appendix.

Reference document

The following documents are available on www.abovsemi.com.

- A34M41x Datasheet
- A34M41x User's manual
  - MPWM functional description
  - ADC functional description

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1. MPWM

1.1 MPWM Introduction

MPWM has a total of three PWM modes that support a variety of application products:

Motor PWM mode for motor control

General purpose PWM mode for various PWM applications

Individual PWM mode for specific PWM applications such as IH cooker

In this document, the Motor PWM mode used in the FOC algorithm is described. For information on the overall features and functions of the MPWM, refer to the user's manual of each MCU.

1.2 MPWM main features in motor PWM mode

Main features of the motor PWM mode are as follows:

Up to six-channel PWM outputs

A 16-bit counter with repeated operation

Up-down count mode

Symmetric / asymmetric modes supported

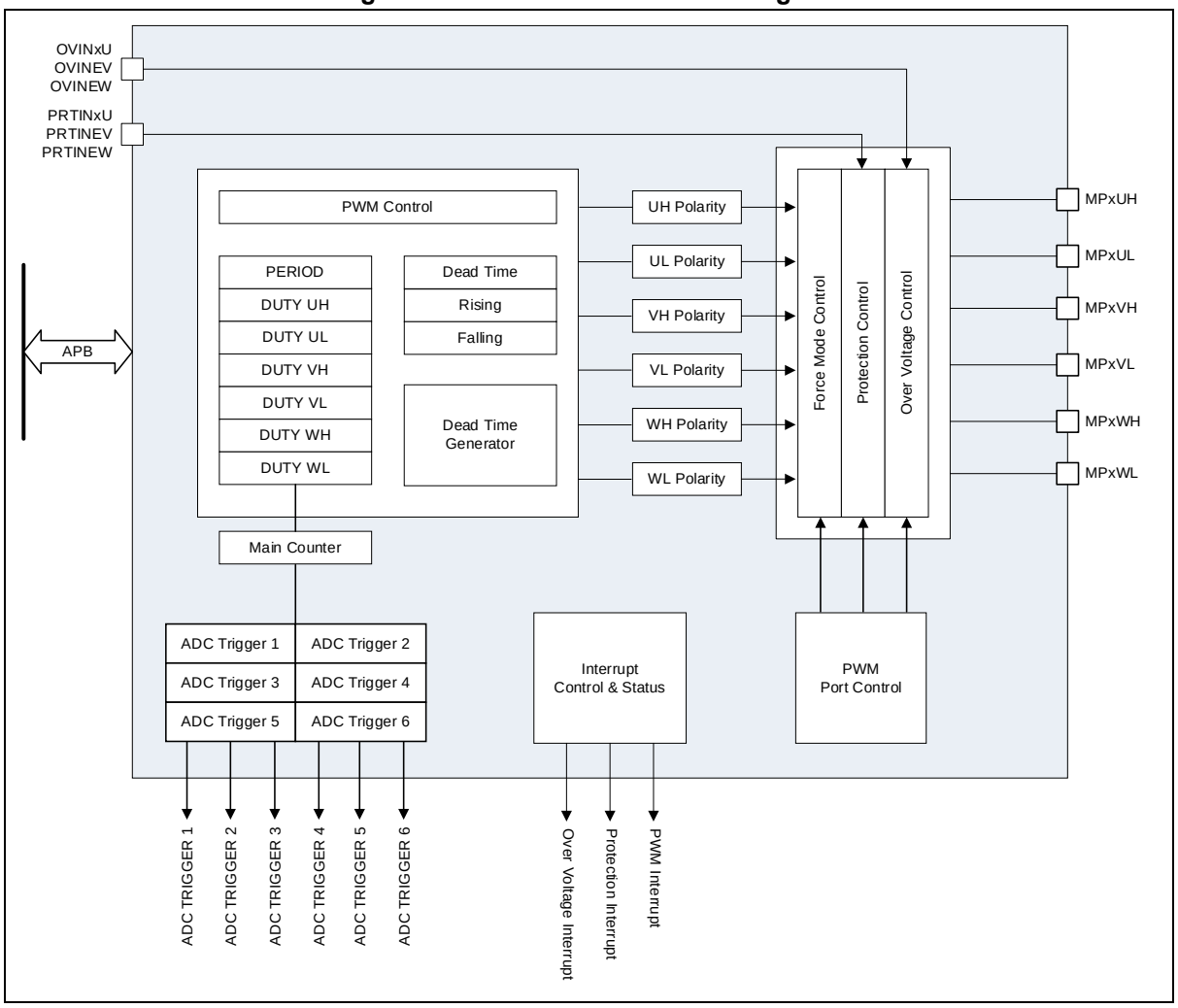
Dead-time insertion supported, to support half-bridge converter circuit

Six ADC trigger outputs

Processing the protection events and overvoltage events

Figure 1 shows a block diagram of the MPWM block in the A34M41x.

Figure 1. A34M41x MPWM block diagram



1.3 MPWM functional description

1.3.1 Symmetric and asymmetric mode

The MPWM has three modes that support various motor applications: 1-ch symmetric mode, 2-ch symmetric mode, and 1-ch asymmetric mode.

Figure 2, Figure 3, and Figure 4 show the timing diagrams of each mode on the A34M41x.

Figure 2. A34M41x 1-ch symmetric timing diagram

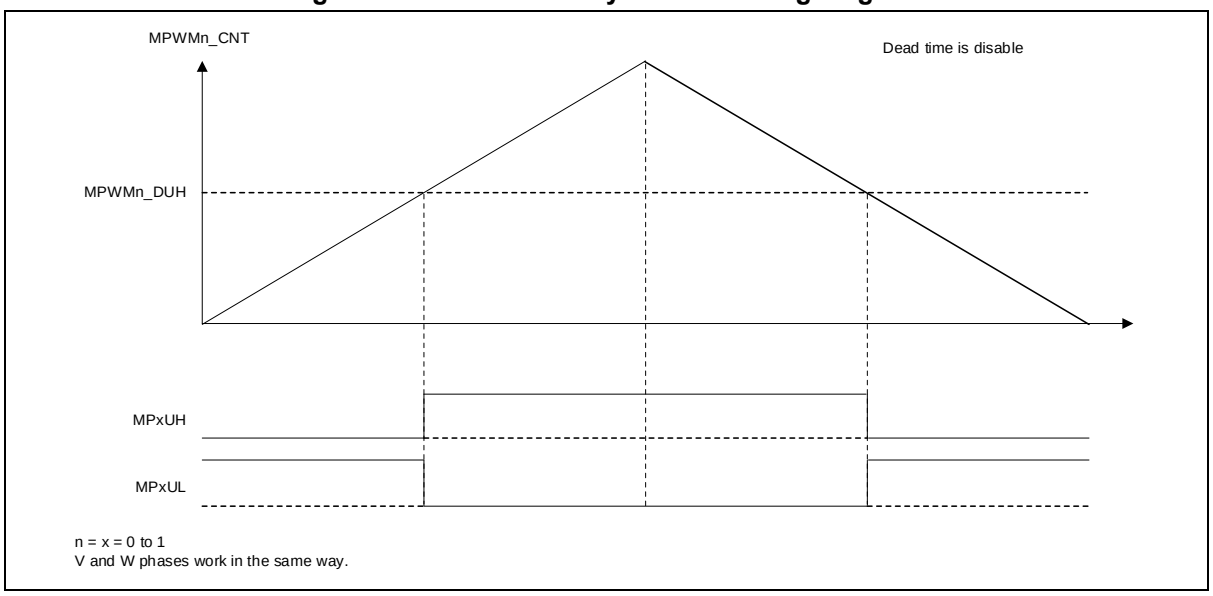


Figure 3. A34M41x 2-ch symmetric timing diagram

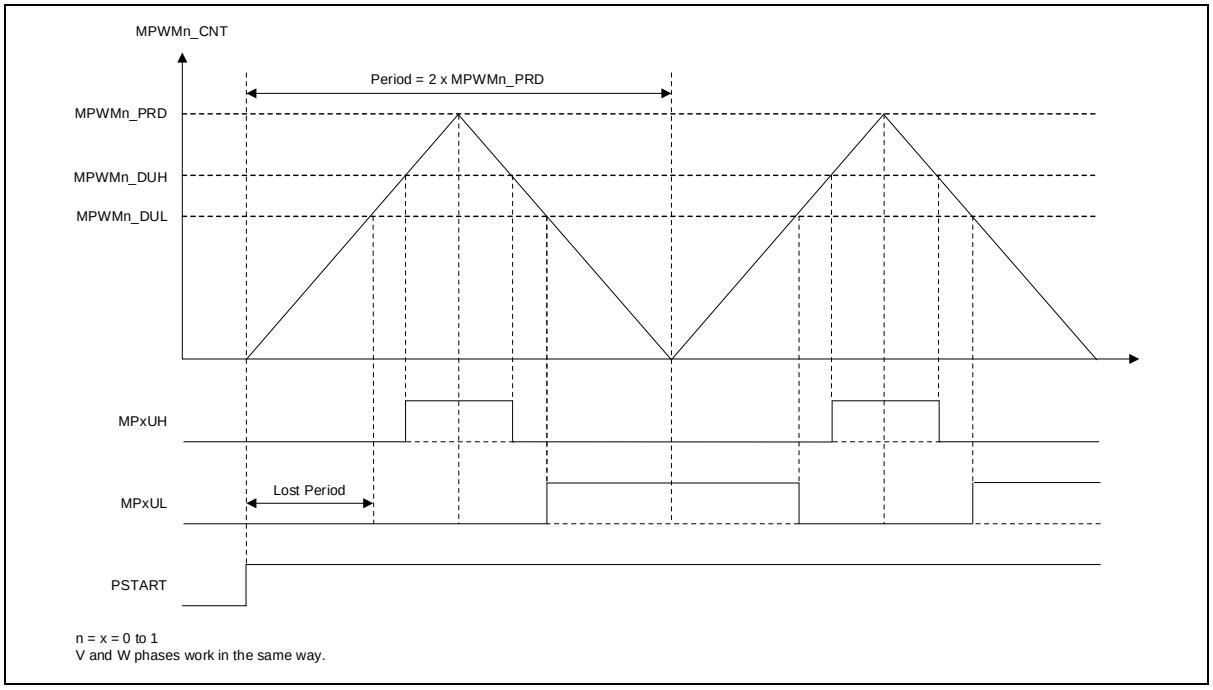
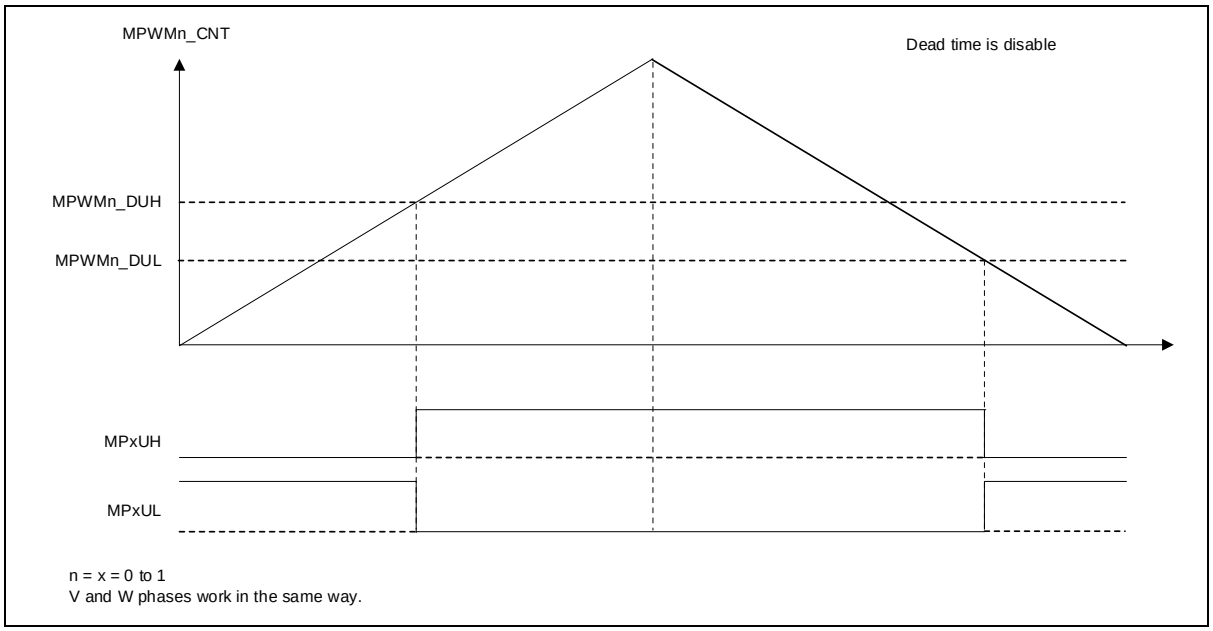


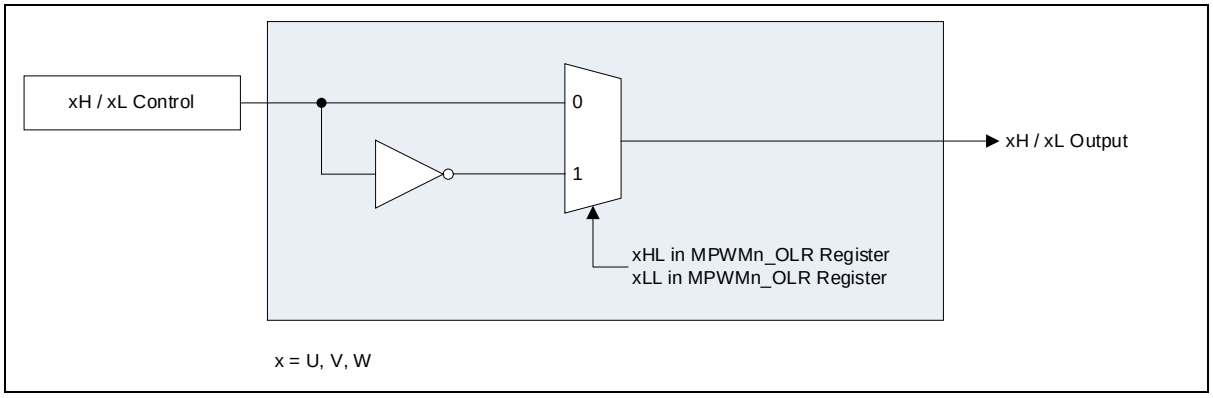
Figure 4. A34M41x 1-ch asymmetric timing diagram



1.3.2 Output control

The MPWM peripheral has six channels consisting of logic blocks as shown in Figure 5, to control the outputs.

Figure 5. Output control block



Based on the logic block configuration, you can use the Output Level Register (OLR) to set the output level both when the counter is not operating and when the counter is operating. The OLR register is set to match the switch polarity of the external circuit and meet the conditions of the PWM form to be implemented.

1.3.3 Complementary outputs and dead-time insertion

Complementary outputs

A control method that enables high side and low side outputs to cross each other in a circuit such as a half-bridge converter is called a complementary symmetric output.

If the function of dead-time insertion (described below in this section) is turned off, the sum of duty cycles on the high side and duty cycles on the low side must be 100 % (period time) in the complementary symmetric output.

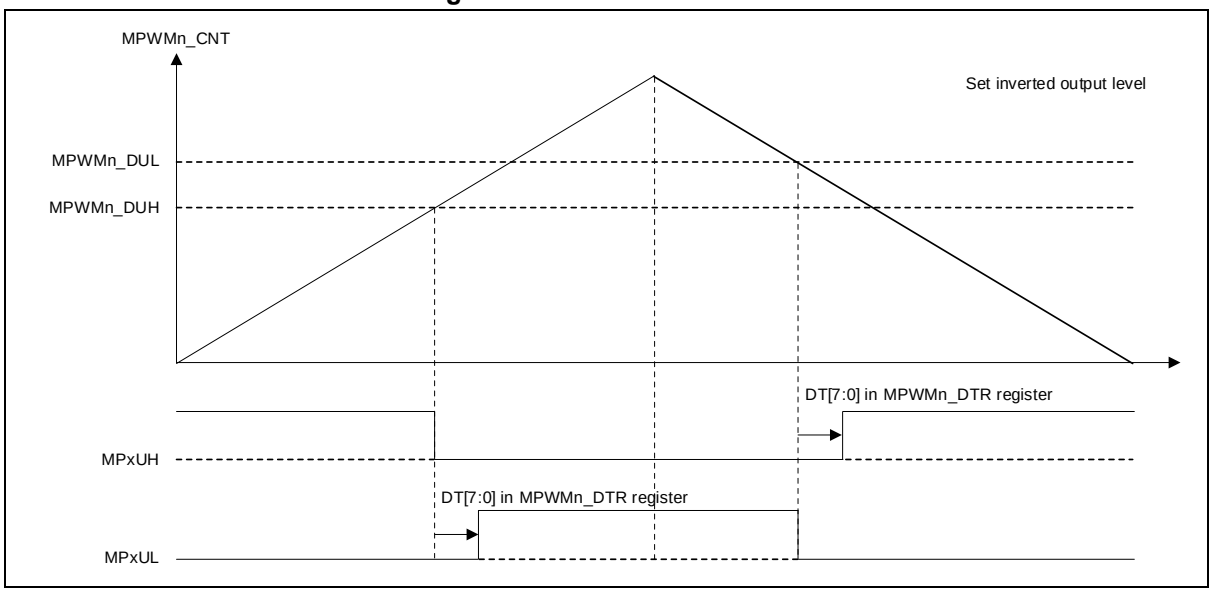
Duty cycle of high side + Duty cycle of low side = 100 %

Dead-time insertion

A delay time, called dead-time, is required in a complementary symmetric circuit. This is to prevent the occurrence of the arm-short circuit due to the switch on / off delay characteristics that is connected to the output.

Figure 6 shows the dead-time operation in 1-ch symmetric mode.

Figure 6. Dead-time insertion



1.3.4 Period / duty update timing

There are several registers that are associated with duty and period, and three phases' outputs need to be changed simultaneously for stable FOC control. Time delays can occur while setting the different registers using software, and the MPWM supports applying duty and period to the MPWM outputs at certain times, to prevent this time delays.

Table 2 shows the timing conditions for updating duty and period.

Table 2. Update timing of duty and period

UAO	TUP	BUP	Description
0	0	1	Duties and periods are updated at every bottom match.
0	1	0	Duties and periods are updated at every period match.
0	1	1	Duties and periods are updated at every bottom and period match.
1	x	x	The duty and period are updated immediately when requested. With this setting, the duty and period registers are updated after two PWM clock cycles.

1.3.5 Protection

Forced output mode

Depending on the application, it can be necessary to immediately stop the external switch operation by setting the PWM output to the high or low level.

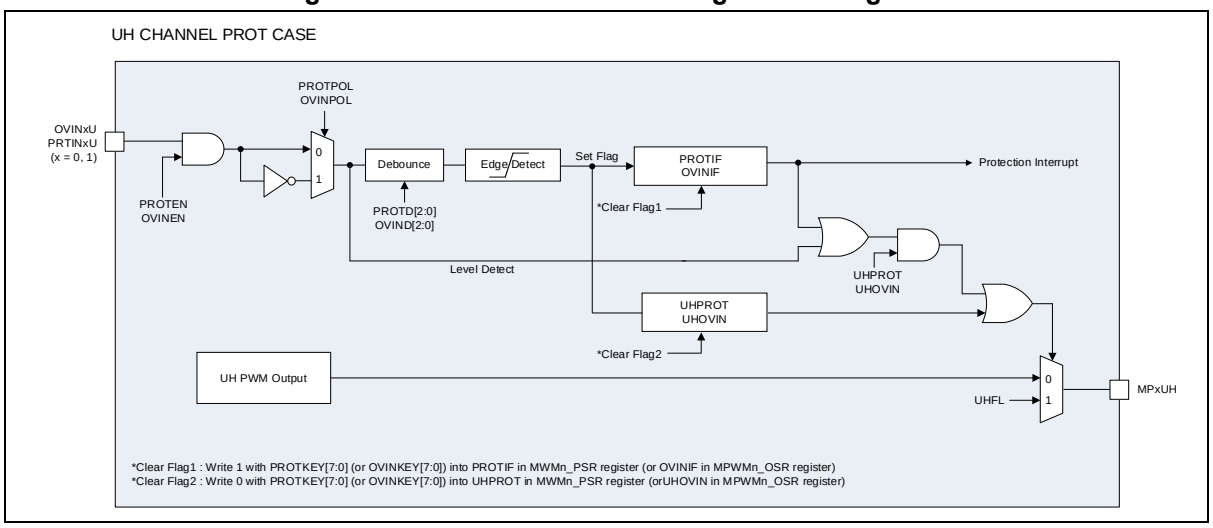
The MPWM can force to control the PWM signal for each MPWM channel, according to the external conditions or user purposes. The MPWM can set the polarities of the six outputs output under the forced condition, and typically, sets each polarity to inactive polarity of the switch connected externally.

The forced output occurs during the protection operation, which is described in the following section.

Protection and overvoltage

Figure 7 is a block diagram for protection and overvoltage. When the protection and overvoltage event occurs, a high or low level signal is input via a corresponding pin to force the predefined high or low level signal output instead of the PWM output.

Figure 7. Protection and overvoltage block diagram



The protection operation sets the following functions:

Enables the protection operation input.

Sets the protection operation input polarity.

Sets the protection operation input debounce.

Sets the protection operation interrupt.

Sets whether to force each output during the protection operation.

The protection operation can set the forced output to enable or disable using the software. However, if you use the software for writing to the Protection Status Register (PSR) or Overvoltage Status Register (OSR), you must write ‘0xCA’ to the PROTKEY bits or ‘0xAC’ to the OVINKEY bits together.

1.3.6 ADC synchronization

The FOC algorithm must be controlled according to the measured data after measuring the current flowing across the shunt resistor in synchronization with PWM. For this, MPWM provides a function that generates the conversion trigger toward the ADC peripheral.

Table 3 lists the trigger sources that the MPWM peripheral can generate.

Table 3. Trigger source by MPWM

Tigger source (ADC peripheral)	Motor PWM mode (A34M41x)
0	ATR1
1	ATR2
2	ATR3
3	ATR4

When the MPWM operates in motor PWM mode, the period matching and bottom trigger are generated at fixed intervals based on the values set in MPWMn_PRD register. In addition, triggers generated by the registers MPWMn_ATRm (m = 1 to 6) may vary according to the ATCNT bits.

The ADC Trigger Register (ATR) can control the following functions:

Selects count match period.

Selects the time to apply to internal logic.

Each ATR register can select the count match period. Using the ATMOD bits, the ADC triggers are generated when the counter and ATCNT bits are matched during the up count / down count / up-down count period. When the counter operates as a up counter or down counter, a single ADC trigger is generated for a period, while two ADC triggers are generated for the period when the counter operates as an up-down counter.

Similar to the registers relating to period and duty, the ATR registers have the function to apply to internal logic at a certain point in time. The FOC algorithm may need to change the ADC trigger timing in conjunction with the PWM duty. For this, the update timing set in UAO / TUP / BUP bits is used.

1.4 MPWM initial setting for FOC

1.4.1 Initialize MPWM peripheral

The code block below shows an example of initializing MPWM and MPWM output ports using the Peripheral Enable Register (SCU_PER2) / Peripheral Clock Enable Register (SCU_PCER2) and Mode Register (MR) of the PCU group.

```
SYST_ACCESS_EN();

SCU->PER2 &= ~(1<<17);
SCU->PCER2 &= ~(1<<17);

SCU->PER2 |= (1<<17);
SCU->PCER2 |= (1<<17);

SCU->PER1 |= (1 << 9)

SYST_ACCESS_DIS();

PORT_ACCESS_EN();

PB->MR1 = 0
      | (3<<20)
      | (3<<16)
      | (3<<12)
      | (3<<8)
      | (3<<4)
      | (3<<0)
      ;

PORT_ACCESS_DIS();
```

1.4.2 Set CR1 and CR2 register of MPWM

The code block below enables the MPWM and stops/ initializes the PWM counter.

```
MPWM1->CR1 = 0
      | (0 << 8) //IRQN
      | (1 << 0) //PWMEN
      ;

MWPM1->CR2 = 0;
```

1.4.3 Set MR register of MPWM

The code block below sets the MPWM to motor PWM mode, 1-ch asymmetric, and up / down counter. In addition, it sets the UAO, TUP, BUP bits to apply the period and duty to the MPWM logic at the point of period match.

```
MPWM1->MR = 0
      | (0 << 14) //MOTORB   0 : motor mode, 1 : normal mode, 3 : individual mode
      | (0 << 7)  //UAO      0 : update are performed at designated timings, 1 : after 2 PWM clocks
      | (1 << 5)  //TUP      0 : not updated, 1 : every period match
      | (0 << 4)  //BUP      0 : not updated, 1 : every bottom match
      | (1 << 1)  //MCHMOD   0 : 2-ch symmetric, 1 : 1-ch asymmetric, 2 : 1-ch symmetric
      | (1 << 0)  //UPDOWN   0 : PWM UP count, 1 : PWM UP/DOWN
      ;
```

1.4.4 Set OLR and FOLR register of MPWM

The code block below sets the MPWM outputs, assuming that the external device characteristic is active high.

```
MPWM1->OLR = 0
      | (0 << 13) // DOLWH
      | (0 << 12) // DOLVH
      | (0 << 11) // DOLUH
      | (0 << 10) // DOLWL
      | (0 << 9)  // DOLVL
      | (0 << 8)  // DOLUL
      | (0 << 5)  // WHL
      | (0 << 4)  // VHL
      | (0 << 3)  // UHL
      | (0 << 2)  // WLL
      | (0 << 1)  // VLL
      | (0 << 0)  // ULL
      ;

MPWM1->FOLR = 0
      | (0 << 5)  // WHFL
      | (0 << 4)  // VHFL
      | (0 << 3)  // UHFL
      | (0 << 2)  // WLFL
      | (0 << 1)  // VLFL
      | (0 << 0)  // ULFL
      ;
```

1.4.5 Set PRD register of MPWM

The code block below sets the PWM period. Due to the characteristics of motor PWM mode in which up / down counter is used, a period is displayed as Figure 2 and calculated as follows:

PRD = ((MPWM clock / (PWM carrier frequency)) / 2 -1)

The code block below calculates the period when the MPWM clock is 60 MHz.

```
MPWM1->PRD = (uint16_t)((60000000UL / (PWM_HZ)) / 2 - 1);
```

1.4.6 Set DTR register of MPWM

The code block below sets the dead-time. The code block below calculates the period when the MPWM clock is 60 MHz.

```
MPWM1->DTR = 0
      | (1 << 15) // DTEN
      | (0 << 14) // PSHRT
      | (0 << 13) // DTMDSEL
      | (0 << 8)   // DTCLK, 0 : CLK/2, 1 : CLK/4, 2 : CLK/8, 3 : CLK/16
      | (0x3C << 0)   // DT (2 usec => 0.000002 * 60000000 / 2 = 60 = 0x3C)
      ;
```

1.4.7 Set PCR / PSR / OCR / OSR register of MPWM

The code block below sets the options for protection. Default values are used for initial settings. (Ignore the key values in the status registers).

```
MPWM1->PCR = 0;
MPWM1->OCR = 0;

MPWM1->PSR = 0;
MPWM1->OSR = 0;
```

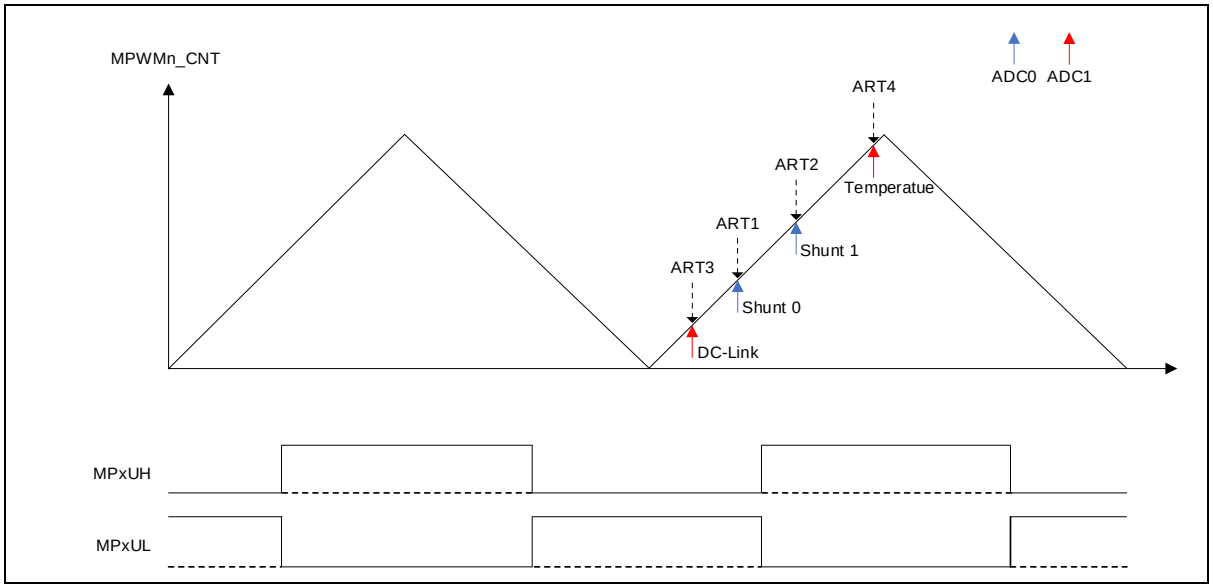
1.4.8 Set ATRn register for default

The code block below is the ADC trigger related initialization code. Additional use of ATR and ‘ATCNT’ setting values can be adjusted according to the application.

```
MPWM1->ATR3 = (1 << 16) | (10);
MPWM1->ATR4 = (1 << 16) | ((MPWM1->PRD - 10) & 0xFFFF);
```

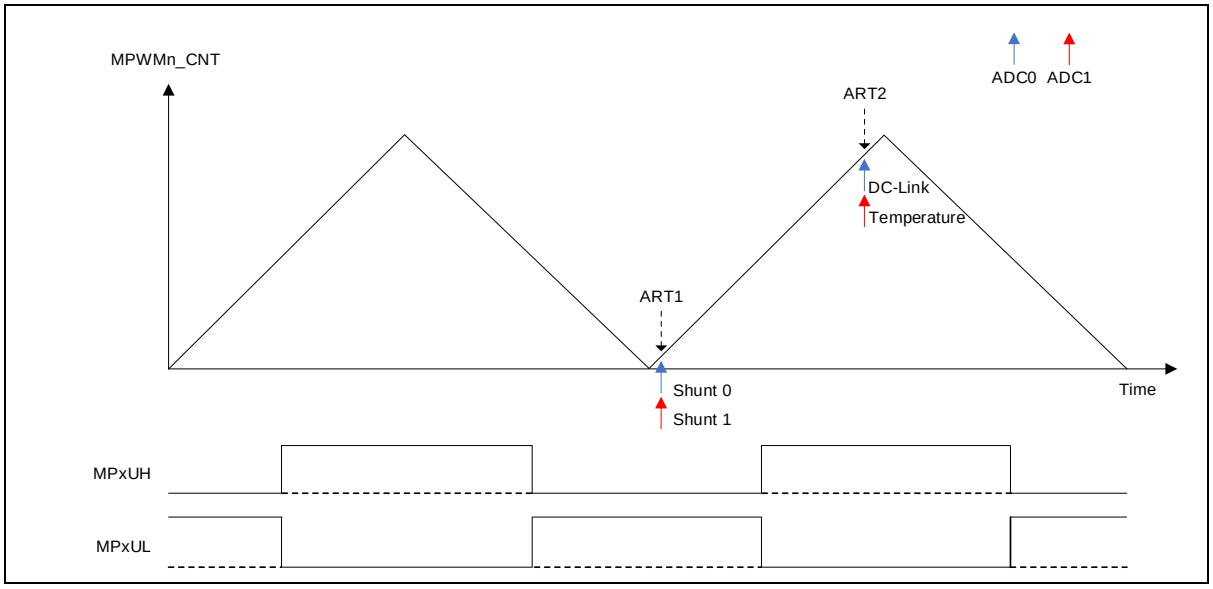
The FOC 1-shunt algorithm uses two ADC units as shown in Figure 8: one measures the current of the motor with the ADC trigger generated at a variable position according to the PWM duty, and the other converts the analog values such as DC link voltage and temperature that are required for motor control.

Figure 8. ADC trigger for FOC 1-shunt



The FOC 3-shunt (or 2-shunt) algorithm uses two ADC units as shown in Figure 9: they convert the analog values representing the motor phase current and DC link voltage and temperature, respectively.

Figure 9. ADC trigger for FOC 3-shunt



1.4.9 Set IER register

When performing the FOC algorithm, the MPWM interrupt timing must be determined to be as close as possible to the point of time when the ADC conversion and FOC algorithm execution begin.

```
1     MPWM1->IER = 0
2               | (1 << 7)   // PRDIEN
3               | (0 << 6)   // BOTIEN
4               | (0 << 5)   // DUHIEN
5               | (0 << 4)   // DULIEN
6               | (0 << 3)   // DVHIEN
7               | (0 << 2)   // DVLIEN
8               | (0 << 1)   // DWHIEN
9               | (0 << 0)   // DWLIEN
10              ;
```

Figure 10. Interrupt timing for FOC 1-shunt

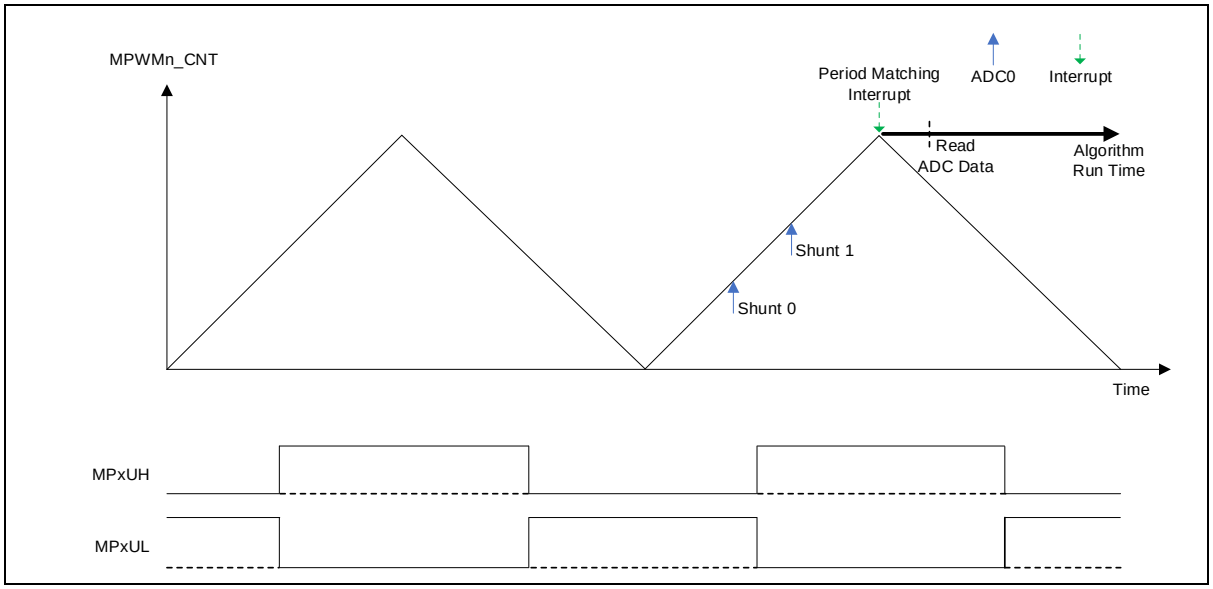
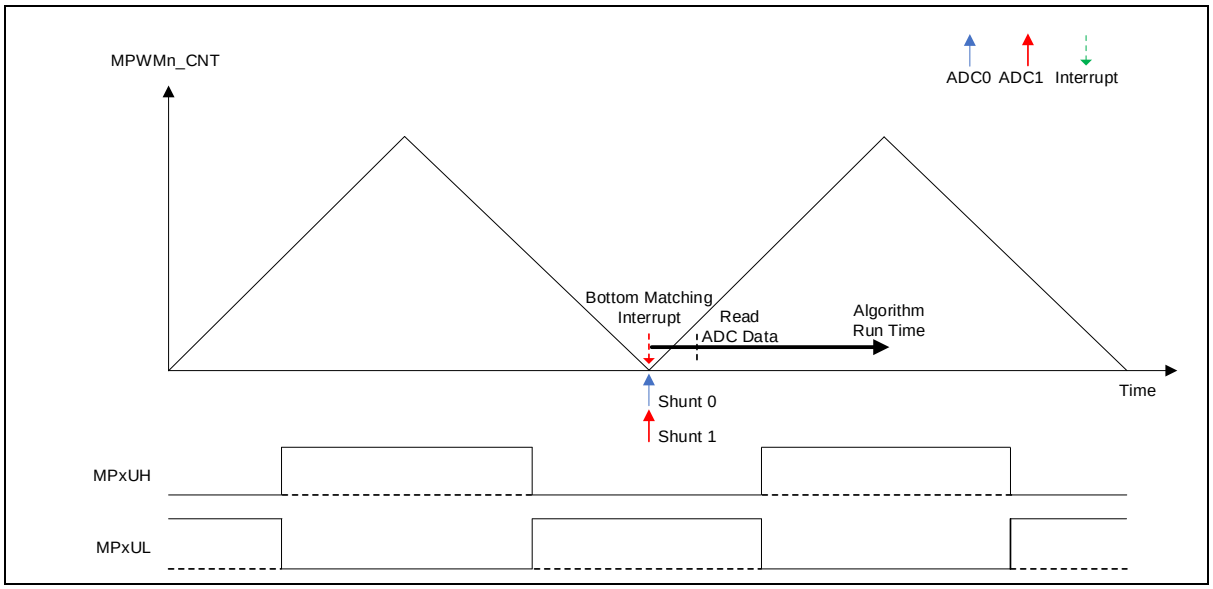


Figure 11. Interrupt timing for FOC 3-shunt



2. ADC

2.1 ADC Introduction

ADC consists of multiple units, and each unit can be controlled individually. The 12-bit ADC allows multiple channels and multiple analog inputs of a channel. The ADC conversions on the multiple channels are performed in single mode, sequential mode, multiple mode, and burst mode. In addition, it triggers in conjunction with TIMER and MPWM.

These features allow the ADC to be used for applications in various fields that require converting analog inputs on the multiple channels at a certain point of time. It is recommended that you use the MPWM trigger to set multiple mode for the FOC algorithm.

This document mainly describes the ADC function used in the FOC algorithm. For more information about the ADC, refer to the A34M41x user's manual.

2.2 ADC main features

Table 4 shows the main ADC features of the A34M41x for the FOC algorithm:

12-bit resolution

Up to 1.5 Msps conversion rate

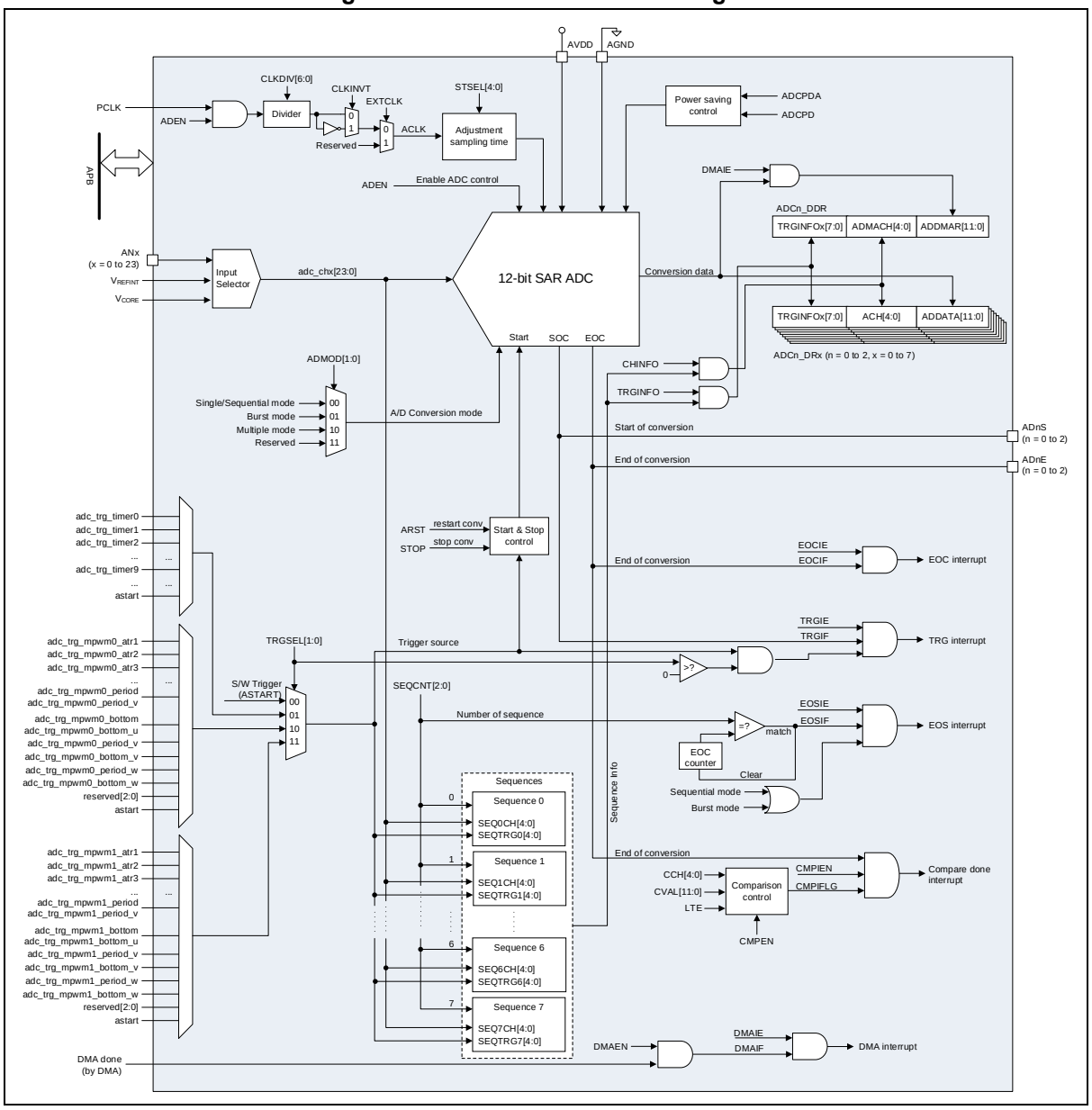
Conversion that works with MPWM triggers

Table 4. ADC spec of A34M41x series

Device name		**A34M418YL (LQFP-120)**	**A34M418VL A34M416VL A34M414VL (LQFP-100)**	**A34M418RL A34M416RL A34M414RL (LQFP-64)**
Unit		3	3	3
Resolution		12-bit	12-bit	12-bit
Conversion rate		1.5 Msps	1.5 Msps	1.5 Msps
Input channel	Total	29	29	21
	External	24	24	16
	Internal	5 (3 reserved)	5 (3 reserved)	5 (3 reserved)

Figure 12 shows a block diagram of A34M41x's ADC.

Figure 12. A34M41x ADC block diagram



2.3 ADC functional description

2.3.1 ADC on-off control (ADCEN)

Before proceeding the ADC conversion, the ADC peripheral must be enabled. Use the ADCEN in ADC_MR register to enable / disable the ADC peripheral, as follows:

1. Software procedure to enable the ADC
   A. Write values to ADC_SR register to initialize the status bit.
   B. Set the ADCEN to '1' to enable the ADC peripheral.
2. Software procedure to disable the ADC
   A. For the ADC conversion, use the ASTOP bit to stop the conversion or wait for the conversion to be completed.
   B. Set the ADCEN to '0' to disable the ADC peripheral.

2.3.2 Channel selection

Assuming that A34M41x is used in this document, each ADC unit consists of 24 channels, separated into the analog inputs from the external GPIO pads and the inputs connected to the inside of the MCU.

In multiple mode of the FOC algorithm, up to eight channels can be selected using SCSR1 / SCSR2 registers.

External analog inputs of MCU

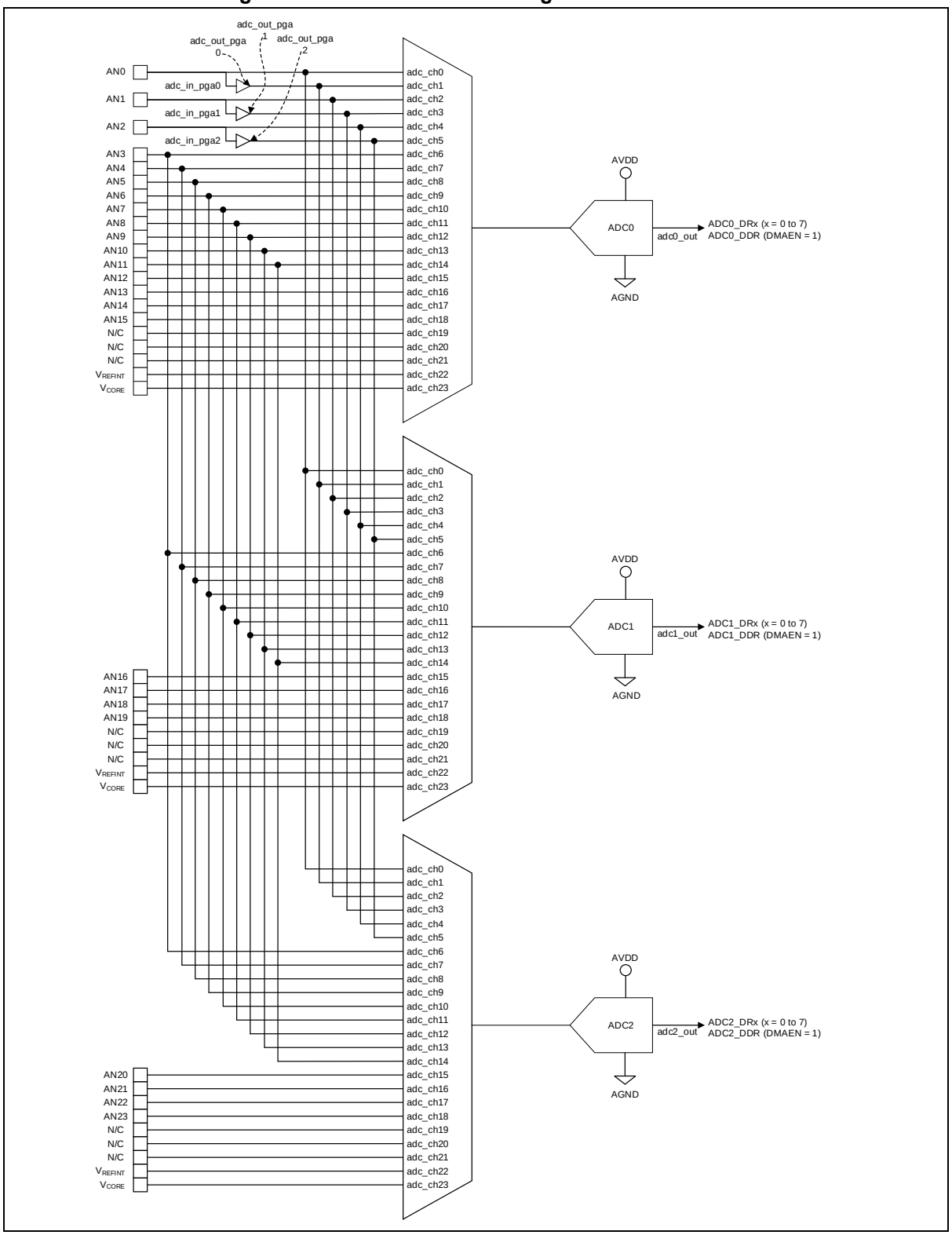
1. Set the alternative function of the port mode to ADC (ANx) using Pn_MR1 or Pn_MR2 (n = A to F) registers, to allow the analog inputs to the ADC channel.
2. Set the pin direction to 'Input' using Pn_CR (n = A to F) registers.
3. Disable the function of internal pull-up and pull-down resistors using Pn_PCR (n = A to F) registers.

Internal analog inputs to MCU

1. Internal 1.0V reference voltage (VREFINT10) is connected to ADC0_AN22, ADC1_AN22, and ADC2_AN22.
2. Internal 1.5V reference voltage (VREFINT15) is connected to ADC0_AN23, ADC1_AN23, and ADC2_AN23.

Figure 13 is a block diagram of showing the internal channel connection of A34M41x ADC.

Figure 13. Internal channel wiring on A34M41x ADC



Multiple mode

Multiple mode conversion performs the ADC conversion according to the order in which the trigger events of the analog signal inputs occur, regardless of the order in which the analog signals input.

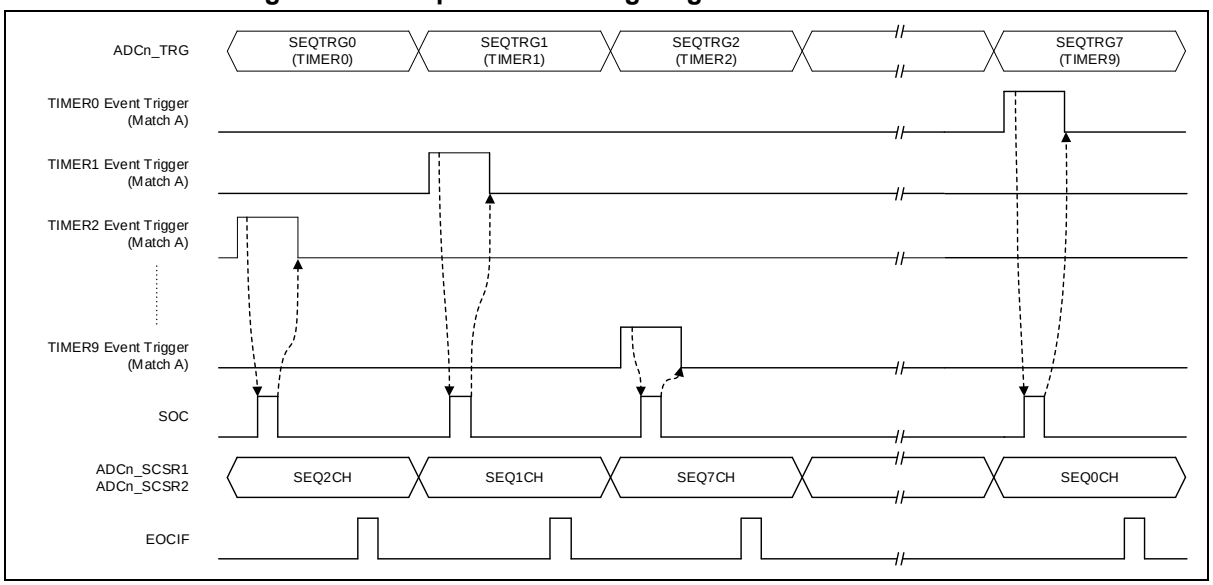
The ADC conversion for analog inputs of the sequence starts when the trigger event occurs, and the converted digital data is stored in ADCx_DRy (x = 0 to 2, y = 0 to 7) registers.

Figure 14 shows a timing diagram of multiple mode that uses eight sequences. Each sequence has a TIMER trigger source that is already set. In the figure, ADC conversion is performed at every trigger event of a sequence, regardless of the sequence order.

After the trigger event occurs in the sequence channel, internal SOC signal is set to ‘H’ until the trigger event terminates. When the trigger event terminates, the SOC signal turns to ‘L’. Then, when the ADC conversion of the analog input signal is completed, End of Conversion (EOC) flag is generated.

Figure 14 shows a multiple mode timing diagram of A34M41x ADC.

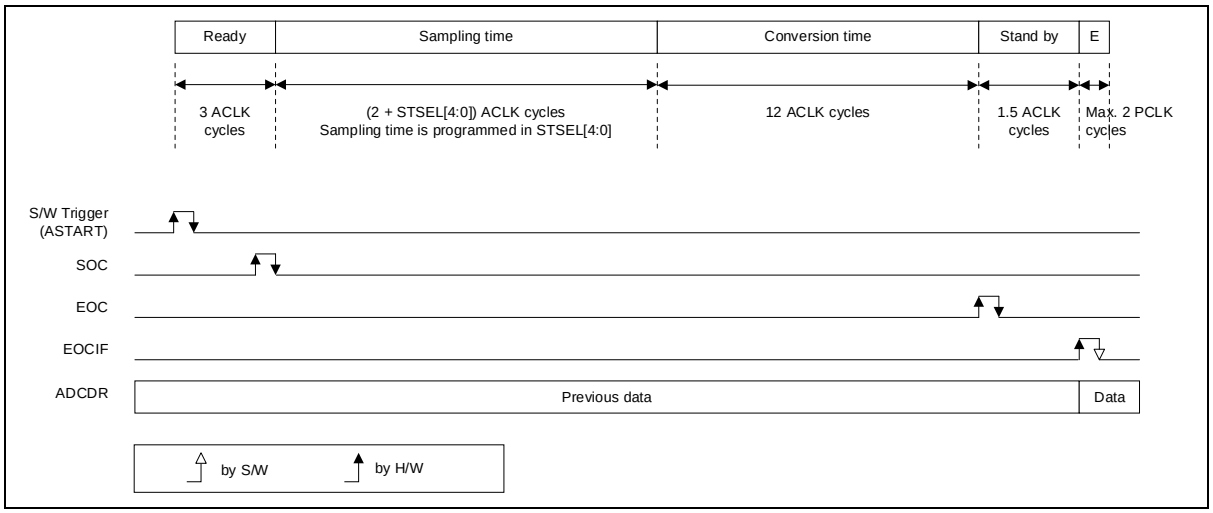
Figure 14. Multiple mode timing diagram of A34M41x ADC



2.3.3 Conversion time

Figure 15 describes the conversion time, assuming that A34M41x is used in this document.

Figure 15. Conversion time of A34M41x ADC



T_{CONV} is calculated as follows:

T_{CONV} = {3 [delay for sync until SOC] + Sampling Time (2 + STSEL[2:0]) + 12 [Conversion Time]

+ 1.5} ACLK cycles [EOC] + max. 2 PCLK cycles [EOCIF]

≒ {Sampling Time (2 + STSEL[4:0]) + 16.5} ACLK cycles

ACLK : ADC clock

PCLK : Peripheral clock

2.3.4 Conversion on external trigger

The ADC conversion or conversion sequence can be triggered by software, TIMER, or external events of MPWM peripheral (Timer captures, input pins, timer interrupts, and so on). The channels of each peripheral can be used as an ADC conversion trigger source. In the FOC algorithm, the MPWM peripheral is used as a trigger source.

ADC trigger sources and TRGSEL bits

Table 5 shows ADC trigger sources that can be selected using TRGSEL[1:0].

Table 5. Configuring ADC external trigger source

TRGSEL[1:0]	Trigger source block	Description
00	ASTART	Disables event triggers and enables the soft trigger only.
01	TIMERx	Enables the timer event trigger and soft trigger.
10	MPWM0	Enables the MPWM0 event trigger and soft trigger.
11	MPWM1	Enables the MPWM1 event trigger and soft trigger.

ADC trigger channel settings (SEQTRGx)

After selecting the trigger source, you can use ADCx_TRG registers to select a single trigger source or up to eight trigger sources in a sequence per ADC unit.

If a trigger source of a single AD channel is used, the trigger source of the SEQTRG0[3:0] is selected. To perform the sequence sampling in burst / multiple / sequential mode, use SEQTRGx (x = 0 to 7) to set appropriate trigger source of the AD conversion channel which is used for each sequence.

The FOC algorithm is set to multiple mode.

Table 6. Trigger source table in A34M41x ADC

Trigger Source (SEQTRGx)	TIMER (TRGSEL[1:0] = 0x3)	MPWM0 (TRGSEL[1:0] = 0x2)	MPWM1 (TRGSEL[1:0] = 0x3)
0	TIMER 0	MP0ATR1	MP1ATR1
1	TIMER 1	MP0ATR2	MP1ATR2
2	TIMER 2	MP0ATR3	MP1ATR3
3	TIMER 3	MP0ATR4	MP1ATR4
4	TIMER 4	MP0ATR5	MP1ATR5
5	TIMER 5	MP0ATR6	MP1ATR6
6	TIMER 6	PERIOD/PERIODU	PERIOD/PERIODU
7	TIMER 7	BOTTOM/BOTTOMU	BOTTOM/BOTTOMU
8	TIMER 8	PERIODV	PERIODV
9	-	BOTTOMV	BOTTOMV
10	-	PERIODW	PERIODW
11	-	BOTTOMW	BOTTOMW
12	-	-	-
13	-	-	-
14	-	-	-
15	ASTART	ASTART	ASTART

ADC trigger interrupt and flag (TRGIE, TRGIF)

Interrupts for the ADC trigger events is set by enabling the TRGIE bit.

TRGIE = 1, Trigger source event interrupt is enabled.

TRGIE = 0, Trigger source event interrupt is disabled.

If the ADC trigger interrupt is enabled, the interrupt TRGIF flag is generated at the point of time when the ADC trigger source event on each ADC conversion channel occurs. The TRGIF flag must be cleared by writing ‘1’.

2.3.5 Data management and information

Data register and data alignment

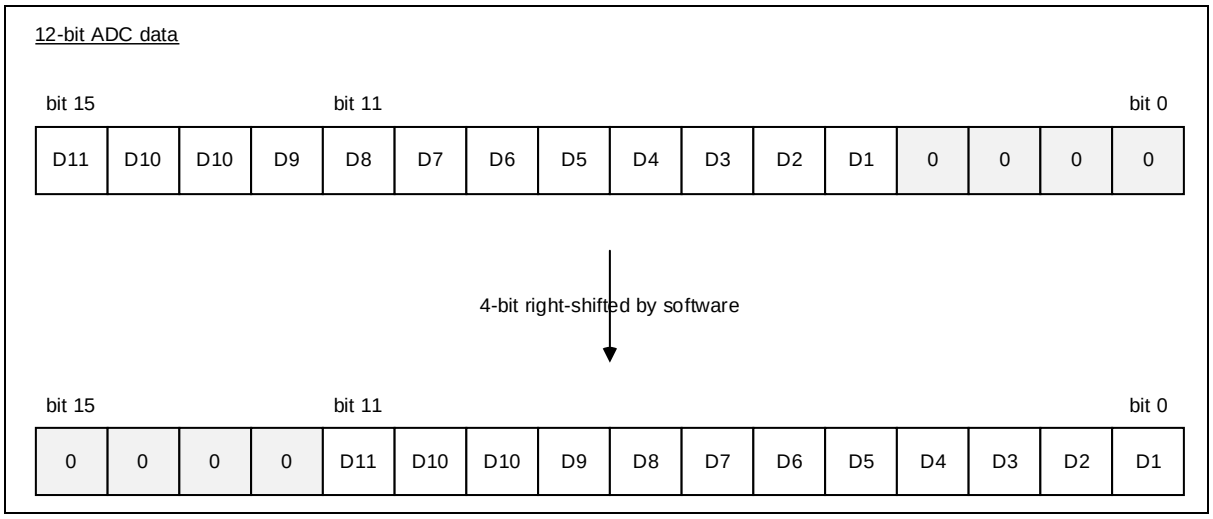
If the EOC event on each ADC conversion channel occurs, the results of the ADC operation are stored in the ADCx_DDR register and the ADCx_DRn (n = 0 to 7) data registers, as left-aligned half-word (16-bit) in the lower 16 bits of the registers.

Figure 16 shows a method of obtaining a 12-bit value by performing the 4-bit right-shifted process using software.

ADCx_DDR : It stores the most recent results of the ADC conversion.

ADCx_DRn (n = 0 to 7): There are eight of these registers representing a sequence. It stores the results of the ADC conversion of the channel assigned to the sequence.

Figure 16. Data align of A34M41x ADC



Channel and trigger information

The upper 16 bits of the ADC data register includes ADC channel information and ADC trigger source information. These bits that control the ADC channel and trigger sources must be set before starting the ADC conversion.

If the CHINFO in ADCx_MR register is set to ‘1’, the ACH[20:16] in ADCx_DR register stores the information of the channel in which data conversion has been conducted.

If the TRGINFO in ADCx_MR register is set to ‘1’, the TRGINFO[31:24] in ADCx_DR register stores the trigger information of the channel in which data conversion has been conducted. The smaller the channel number specified in a sequence, the higher the priority. The TRGINFO[31:24] shows the channel to execute first.

Figure 17 shows that the channels (from adc_ch0 to adc_ch7) are assigned in SEQxCH (x = 0 to 7) sequences, and then the channel and trigger information are found in the ADCx_(D)DR register if CHINFO = 1 and TRGINFO = 1.

Figure 17. Channel and trigger source information of ADC sequence

	SEQ0CH	SEQ1CH	SEQ2CH	SEQ3CH	SEQ4CH	SEQ5CH	SEQ6CH	SEQ7CH
	adc_ch0	adc_ch1	adc_ch2	adc_ch3	adc_ch4	adc_ch5	adc_ch6	adc_ch7
ACH[4:0] (Channel number of AD conversion)	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
TRGINFO[7:0] (Trigger source priority of AD conversion)	0x01	0x02	0x04	0x08	0x10	0x20	0x40	0x80

2.4 ADC initial setting for FOC

2.4.1 Initialize ADC peripheral

The code block below initializes ADC0 and ADC1 peripherals using Peripheral Enable Register (SCU_PER2) / Peripheral Clock Enable Register (SCU_PCER2) and Mode Register (MR) of the PCU group.

```
1   // ADC0 : Trigger Mode
2   SYST_ACCESS_EN();
3   SCU->PER2 &= ~(1 << 20);
4   SCU->PCER2 &= ~(1 << 20);
5
6   SCU->PER2 |= (1 << 20);
7   SCU->PCER2 |= (1 << 20);
8   SYST_ACCESS_DIS();
9
10  // ADC1 : Single Mode
11  SYST_ACCESS_EN();
12  SCU->PER2 &= ~(1 << 21);
13  SCU->PCER2 &= ~(1 << 21);
14
15  SCU->PER2 |= (1 << 21);
16  SCU->PCER2 |= (1 << 21);
17  SYST_ACCESS_DIS();
```

2.4.2 Set MR register of ADC

The code block below initializes the mode settings. Both ADC0 and ADC1 peripherals are set to multiple mode and use two triggers for ADC conversion operation.

For information on the MPWM ADC trigger settings, see section 1.4.8.

```
ADC0->MR = 0
        | (0 << 21) // TRGINFO
        | (0 << 20) // CHINFO
        | (0 << 17) // DMAEN
        | (A34M41x_ADC_STSEL << 12)
        | (1 << 8)   // SEQCNT
        | (1 << 7)   // ADCEN
        | (1 << 6)   // ARST
        | (0 << 4)   // ADCMOD
        | (3 << 0)   // TRGSRC
        ;
ADC1->MR = 0
        | (0 << 21) // TRGINFO
        | (0 << 20) // CHINFO
        | (0 << 17) // DMAEN
        | (A34M41x_ADC_STSEL << 12)
        | (1 << 8)   // CEQCNT
        | (1 << 7)   // ADCEN
        | (1 << 6)   // ARST
        | (0 << 4)   // ADCMOD
        | (3 << 0)   // TRGSRC
        ;
```

2.4.3 Set CSCR register of ADC

The code block below initializes current sequence channel registers. Both ADC0 and ADC1 are explicitly initialized to ‘0’.

```
ADC0->CSCR = 0x00;

ADC1->CSCR = 0x00;
```

2.4.4 Set CCR register of ADC

The code block below initializes clock control registers. Both ADC0 and ADC1 are set to divide internal clock (PCLK) into five to set the ADC clock (ACLK) to 24 MHz. (A34M41x_ADC_CLKDIV = 5)

```
ADC0->CCR = 0
        | (A34M41x_ADC_CLKDIV << 8)   // CLKDIV
        | (0 << 7)   // ADCPD
        | (0 << 6)   // EXTCLK
        | (0 << 5)   // CLKINVT
        ;

ADC1->CCR = 0
        | (A34M41x_ADC_CLKDIV << 8)   // CLKDIV
        | (0 << 7)   // ADCPD
        | (0 << 6)   // EXTCLK
        | (0 << 5)   // CLKINVT
        ;
```

2.4.5 Set TRG register of ADC

The code block below initializes trigger select registers.

The FOC 1-shunt algorithm uses different ADCs for shunt current measurements and other analog signal measurements. Accordingly, the ADC0 is set to receive the trigger in MPWM1_ATR1 and MPWM1_ATR2 registers while the ADC1 is set to receive the trigger in MPWM1_ATR3 and MPWM1_ATR4 registers.

The trigger timing depends on the algorithm. See section 1.4.8 for information on the MPWM ATR settings.

```
1   ADC0->TRG = 0
2           | (0 << 28) // SEQTRG7
3           | (0 << 24) // SEQTRG6
4           | (0 << 20) // SEQTRG5
5           | (0 << 16) // SEQTRG4
6           | (0 << 12) // SEQTRG3
7           | (0 << 8)   // SEQTRG2
8           | (1 << 4)   // SEQTRG1 (MP1ATR2)
9           | (0 << 0)   // SEQTRG0 (MP1ATR1)
10          ;
11
12  ADC1->TRG = 0
13          | (0 << 28) // SEQTRG7
14          | (0 << 24) // SEQTRG6
15          | (0 << 20) // SEQTRG5
16          | (0 << 16) // SEQTRG4
17          | (0 << 12) // SEQTRG3
18          | (0 << 8)   // SEQTRG2
19          | (3 << 4)   // SEQTRG1 (MP1ATR4)
20          | (2 << 0)   // SEQTRG0 (MP1ATR3)
21          ;
```

2.4.6 Set SCSR1 / SCSR2 register of ADC

The code block below sets the channel on which the ADC conversion is conducted when the trigger is generated.

The FOC 1-shunt algorithm sets the ADC0 to convert the I_COM current twice per period for the phase-current reconstruction and ADC1 to convert the temperature and DC-link voltage.

The trigger timing depends on the algorithm. See section 1.4.8 for information on the MPWM ATR register settings.

```
ADC0->SCSR1 = 0
            | (0 << 24) // SEQ4CH
            | (0 << 16) // SEQ3CH
            | (7 << 8)   // SEQ2CH (AN4, I_COM)
            | (7 << 0)   // SEQ1CH (AN4, I_COM)
            ;

ADC0->SCSR2 = 0;

ADC1->SCSR1 = 0
            | (0 << 24) // SEQ4CH
            | (0 << 16) // SEQ3CH
            | (8 << 8)   // SEQ2CH (AN5, IPM_Temp)
            | (0 << 0)   // SEQ1CH (AN0, VDC_LINK)
            ;

ADC1->SCSR2 = 0;
```

2.4.7 Set CR / SR register of ADC

The code block below initializes ADC control registers and status registers.

```
ADC0->CR = 0
            | (0 << 7)   // ASTOP ADC conversion stop
            | (0 << 1)   // TRGCLR
            | (0 << 0)   // ASTART
            ;

ADC0->SR    = 0xFFFF;

ADC1->CR = 0
            | (0 << 7)   // ASTOP ADC conversion stop
            | (0 << 1)   // TRGCLR
            | (0 << 0)   // ASTART
            ;

ADC1->SR    = 0xFFFF;
```

2.4.8 Set IER register of ADC

The code block below initializes ADC interrupt enable registers.

The FOC algorithm operates based on the MPWM interrupts and does not use the ADC interrupts. However, depending on the application, the ADC EOS or EOC interrupts can be used instead of using the MPWM.

```
1       ADC0->IER = 0
2                 | (0 << 4)   // DMAIE
3                 | (0 << 3)   // TRGIE
4                 | (0 << 2)   // EOSIE
5                 | (0 << 0)   // EOCIE
6                 ;
7
8       ADC1->IER = 0
9                 | (0 << 4)   // DMAIE
10                | (0 << 3)   // TRGIE
11                | (0 << 2)   // EOSIE
12                | (0 << 0)   // EOCIE
13                ;
```

Appendix. Device overview

Table 7. A34M41x series features

Part No.	A34M41x
CPU Core	Cortex-M4F
Clock Speed	120 MHz
Flash/SRAM	512+32 KB / 64 KB
Timer	16-bit x 10 32-bit FRT x 2 32-bit WDT
DMAC	16-Ch
UART / SPI / I2C	6 / 3 / 2
3P Motor PWM	2
GPIO	107
Special Functions	CAN 2.0B QEI x 2 PGA x 3 COMP x 4
Others	IEC60730 CRC16 AES-128
ADC	12-bit 1.5 Msps
Internal OSC	32 MHz (±3 %)
Analog IP	POR / VDC / LVD
Op. Temp.	-40 ℃ to +85 ℃
Op. Voltage	2.5 V to 5.5 V
PKG	LQFP-120 LQFP-100 LQFP-64
Status	MP
Applications	System Aircon High speed FOC IoT App.

Revision history

Revision	Date	Notes
1.00	Jun. 7, 2021	Initial release.
1.01	Nov. 29, 2022	Updated the template of this document.
1.02	Dec. 2, 2024	Updated the disclaimer.

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